Ali Abbas

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Education

University of British Columbia September 2023 – April 2028
BSc. Data Science - Outstanding International Student Award *Vancouver, BC*

Technical Experience

Data Engineer | ESGTree June 2026 – Dec 2026

- Engineered an **ETL pipeline** integrating four Canadian public data sources (AAFC satellite crop-inventory rasters, CanSIS national soil polygons, Statistics Canada yield tables, and the federal **Holos** whole-farm emissions model) into a normalized field-level schema powering all downstream cost and emissions estimates.
- Built the **geospatial** layer of the pipeline: derived multi-year field cropping histories from **30m satellite rasters** by querying an ImageServer histogram endpoint instead of downloading imagery, joined soil polygons to field geometries, and ran **Sentinel-2 NDVI** time-series analysis in **Google Earth Engine** to detect ground-cover practices.
- Shipped a reusable **Python package** computing crop-rotation nitrogen credits and regulatory compliance verdicts from field-level cropping history, benchmarked against published provincial agronomic and cost-of-production data.
- Attached a **PCAF 1–5 data-quality score** to every estimate, sharpening as farmer-supplied data is ingested, and delivered a **lender portfolio dashboard** (**React**, **Leaflet**) mapping 75+ farm operations by emissions intensity with per-farm drill-down and **XLSX** export.
- Expanding scope from Sept. 2026 to the platform's **financed emissions** module for a Desjardins pilot deployment, enabling commercial borrowers to self-report operational emissions for **Scope 3 Category 15** disclosure.

Data & Signals Developer | *UBC MINT Engineering Design Team* Sep 2024 – Present

- Built **EEG/EMG signal-processing pipelines** in **Python** and **SciPy** (bandpass and notch filtering, RMS envelope extraction, MVC normalization) feeding downstream gesture classification for a robotic arm.
- Consolidated duplicated EEG and EMG data-validation scripts into a single **quality-control module** detecting flatlined channels, amplitude railing, DC drift, and mains interference via **Welch PSD**.
- Built a real-time **EEG streaming visualization** platform (**Next.js**, **WebSockets**) ingesting **256 messages/sec**, plus a bidirectional **CSV import/export** system enabling offline analysis and reproducible runs.

Data Engineering & Analytics Projects

Engine Fault Detection Pipeline | *Python, Pandas, Scikit-Learn, AWS S3, Streamlit, Docker*

- Built an **end-to-end ML pipeline** processing NASA C-MAPSS turbofan sensor data through schema validation, rolling-window feature engineering, and Random Forest fault classification, with versioned model artifacts persisted to **AWS S3** via a swappable storage layer.
- Used engine-grouped cross-validation to prevent data leakage between consecutive sensor readings from the same unit, achieving **0.88 F1** and **0.93 recall** on held-out engines.
- Deployed an interactive **Streamlit dashboard** ranking engines by predicted failure risk, backed by a containerized pipeline (**Docker**) with automated tests covering the no-leak split guarantee.

Spotify Chart Sustainability Analysis | *Python, Oracle SQL, MongoDB, Scikit-Learn*

- Designed and built a multi-source **ETL pipeline** processing **30,000+** records across Spotify Audio Features, Billboard Hot 100, and Charts datasets, resolving entity-resolution conflicts across mismatched schemas.
- Implemented parallel relational (**Oracle SQL**) and document (**MongoDB**) schemas and benchmarked query performance across the different implementations.
- Built a multivariate **linear regression** model (**Scikit-Learn**) predicting chart longevity from audio features, applying hypothesis testing to isolate the audio features with statistically significant predictive power.

Technical Skills

Languages: Python, SQL, R, JavaScript/TypeScript, Java
Data & ML: Pandas, NumPy, SciPy, Scikit-Learn, ETL pipeline design, data modelling, schema design (relational & document), feature engineering, regression, classification, cross-validation, hypothesis testing, model evaluation
Databases & Tools: PostgreSQL, Oracle SQL, MongoDB, Google Earth Engine, Docker, AWS S3, Streamlit, Git, REST, WebSockets